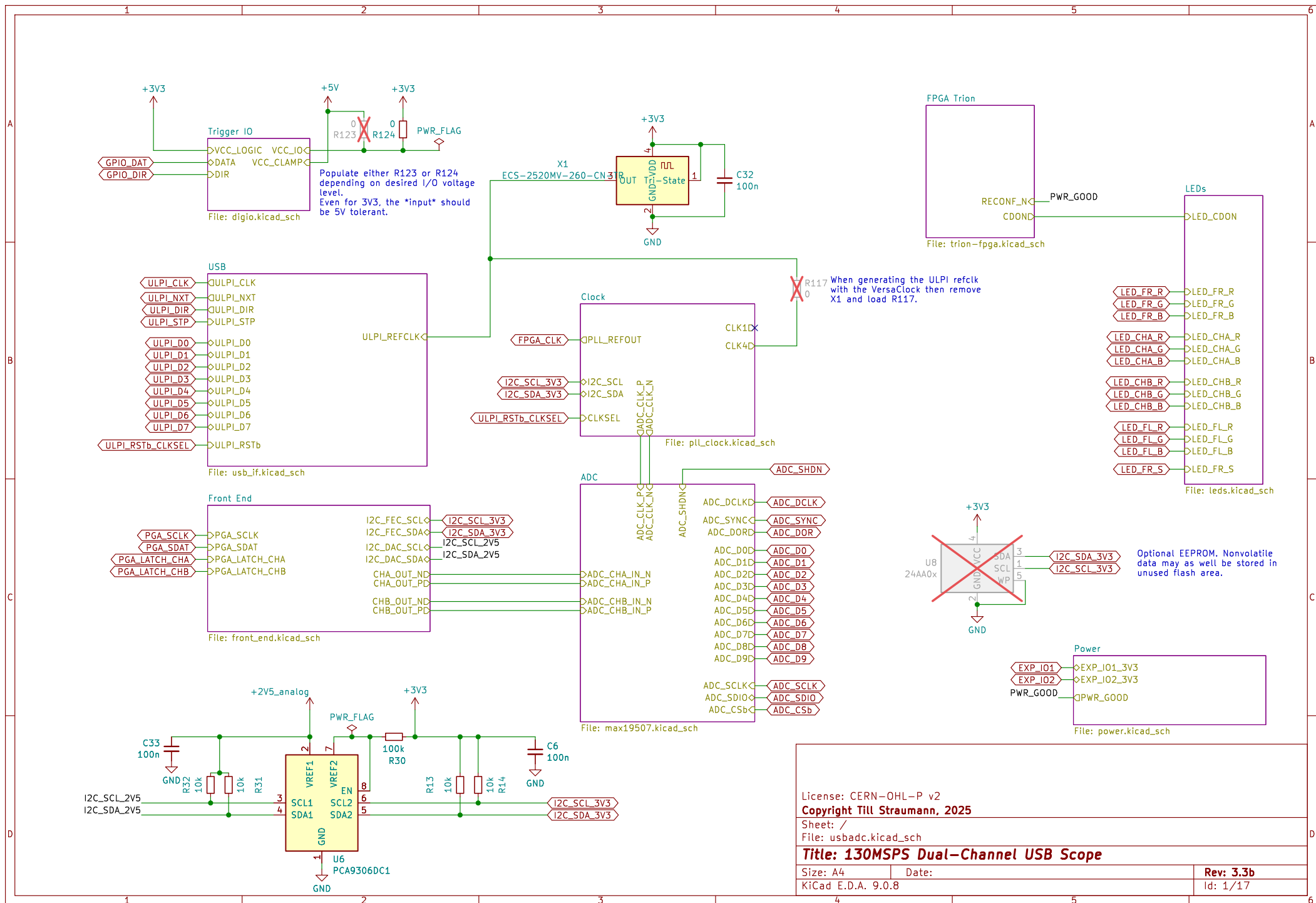
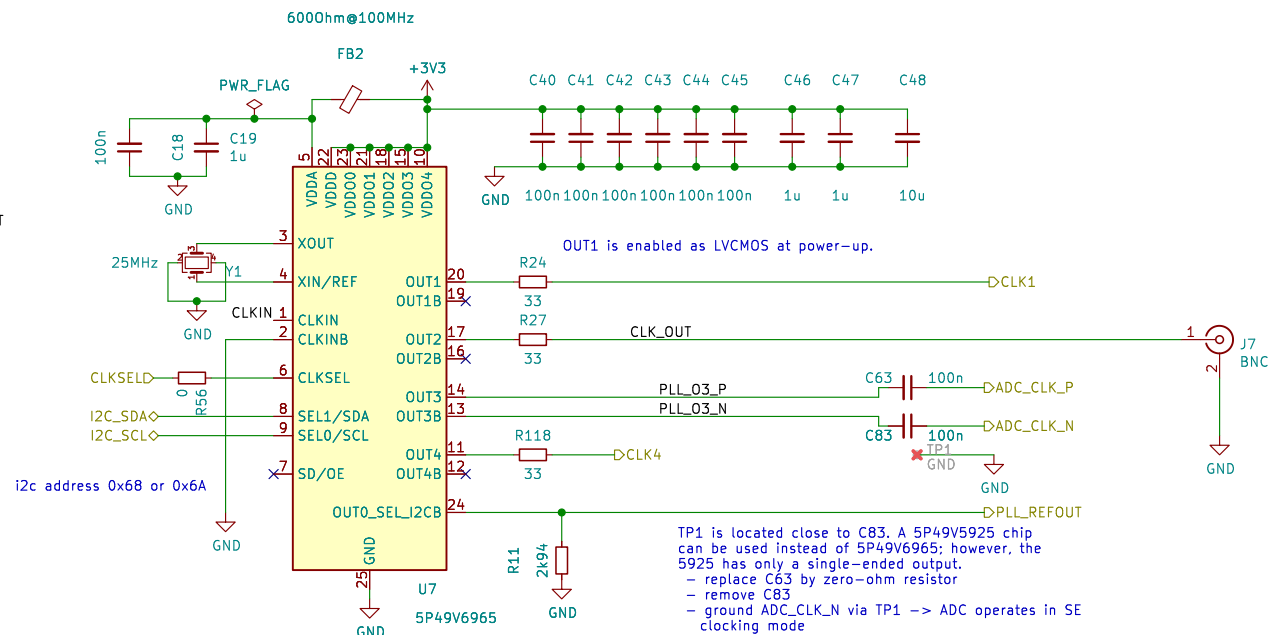




When OUT2 is disabled and CLKSEL switched then an external reference can be fed from the BNC (e.g., lock this scope to a second/master unit for 4 phase-locked ADC channels).



TP1 is located close to C83. A 5P49V5925 chip can be used instead of 5P49V6965; however, the 5925 has only a single-ended output.

- replace C63 by zero-ohm resistor
- remove C83
- ground ADC_CLK_N via TP1 -> ADC operates in SE clocking mode

Another alternative is the 5P49V6975 which is a 6965 with a built-in crystal. DNL Y1 if you load such a device.

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Sheet: /Clock/
File: pll_clock.kicad_sch

Title: Clock Generator

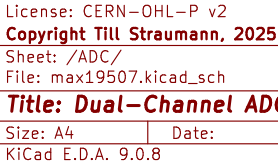
Size: A4

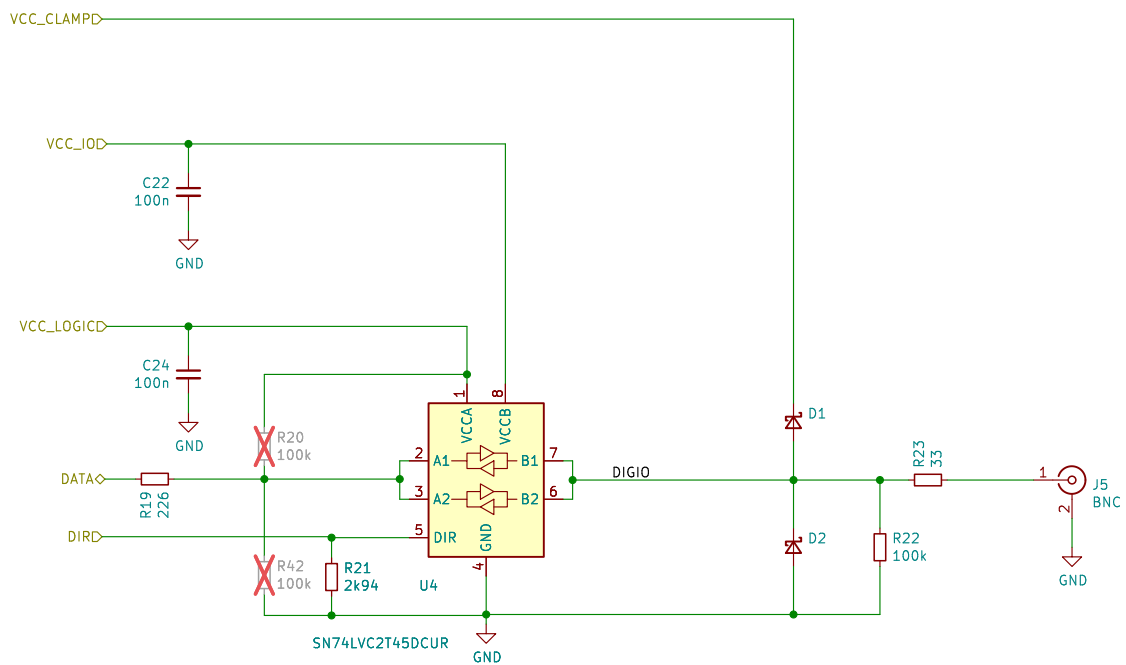
Date:

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Rev: 3.3b

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(LVCMOS18) min pullup: 8.4k; max. pulldown: 3.3k
 2t45 VLmax: 0.35Vcc -> pulldown < 4.5k

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Sheet: /Trigger IO/
 File: digio.kicad_sch

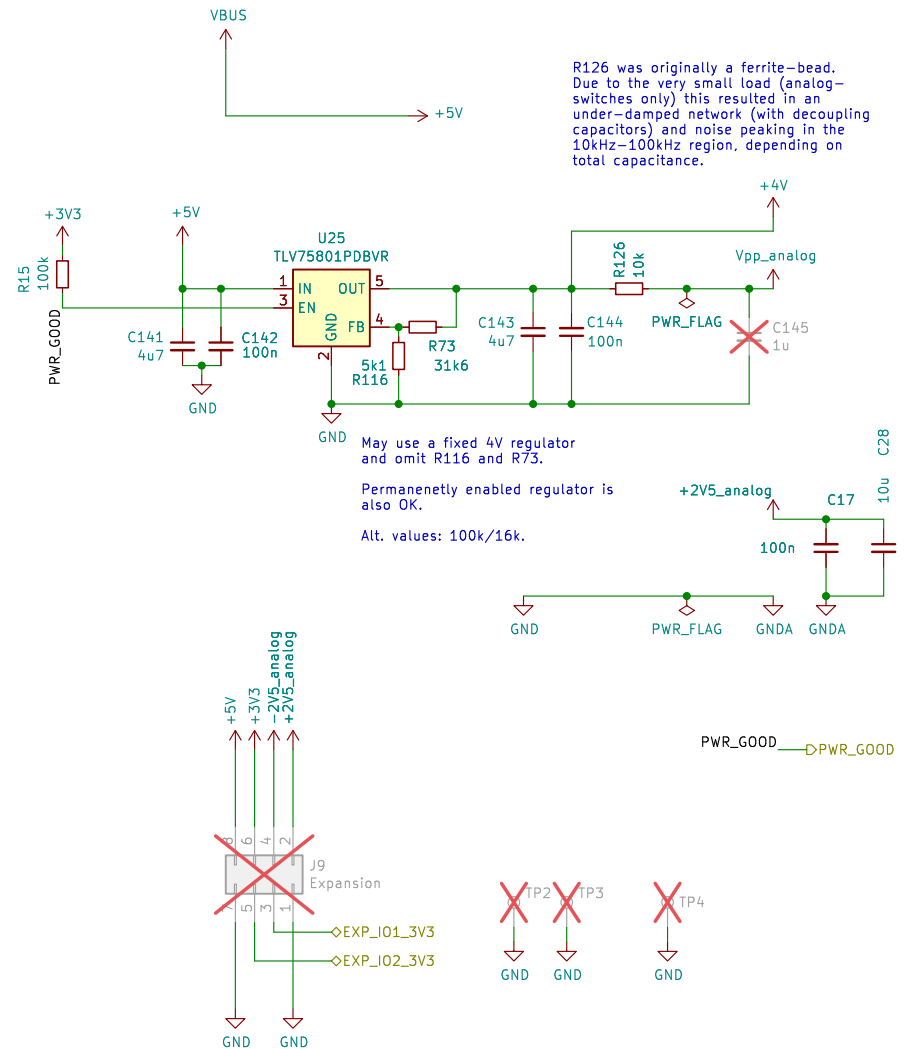
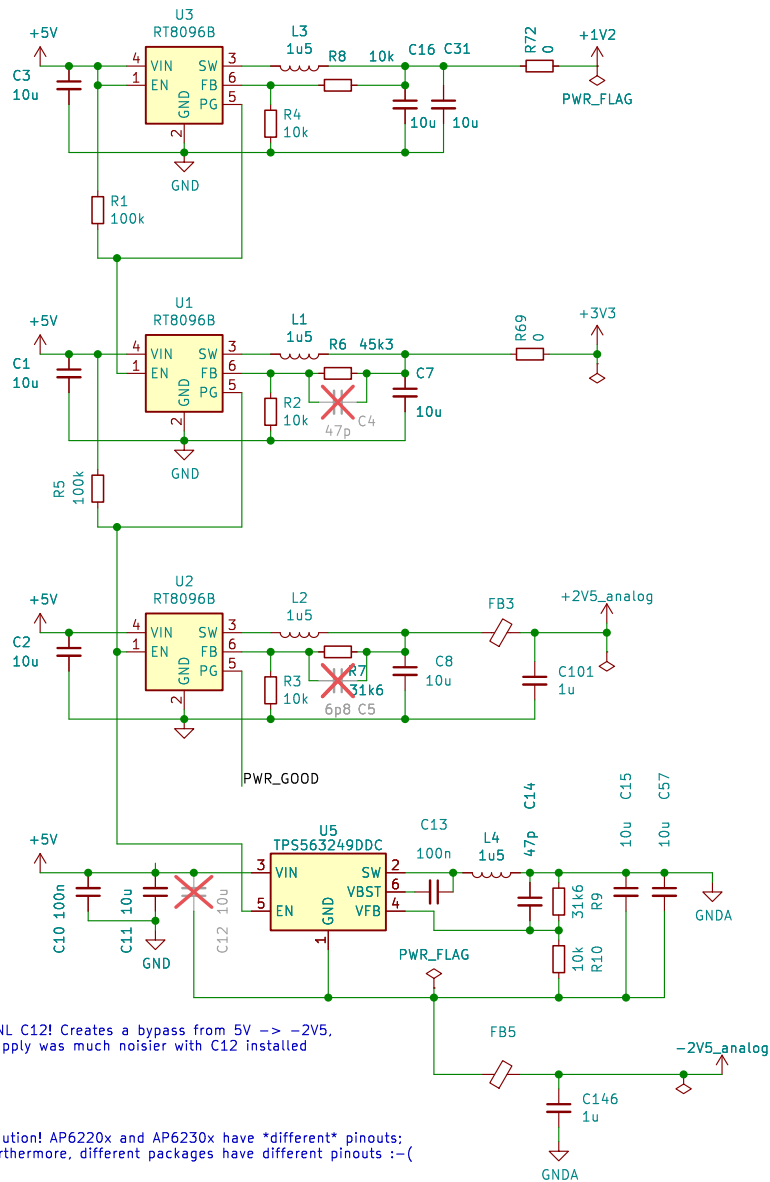
Title: GPIO (External Trigger In/Out)

Size: A4
 KiCad E.D.A. 9.0.8

Date:

Rev: 3.3b

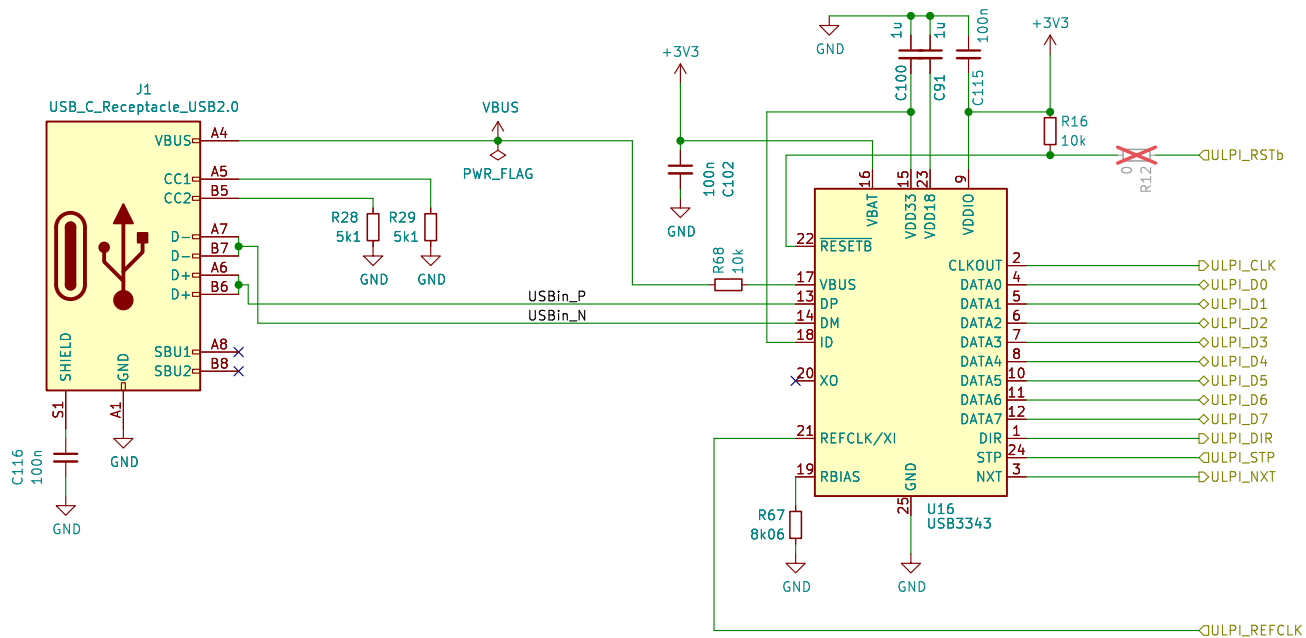
Id: 7/17



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 Sheet: /Power/
 File: power.kicad_sch

Title: Power Supply and Expansion Connector

Size: A4	Date:	Rev: 3.3b
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Sheet: /USB/
 File: usb_if.kicad_sch

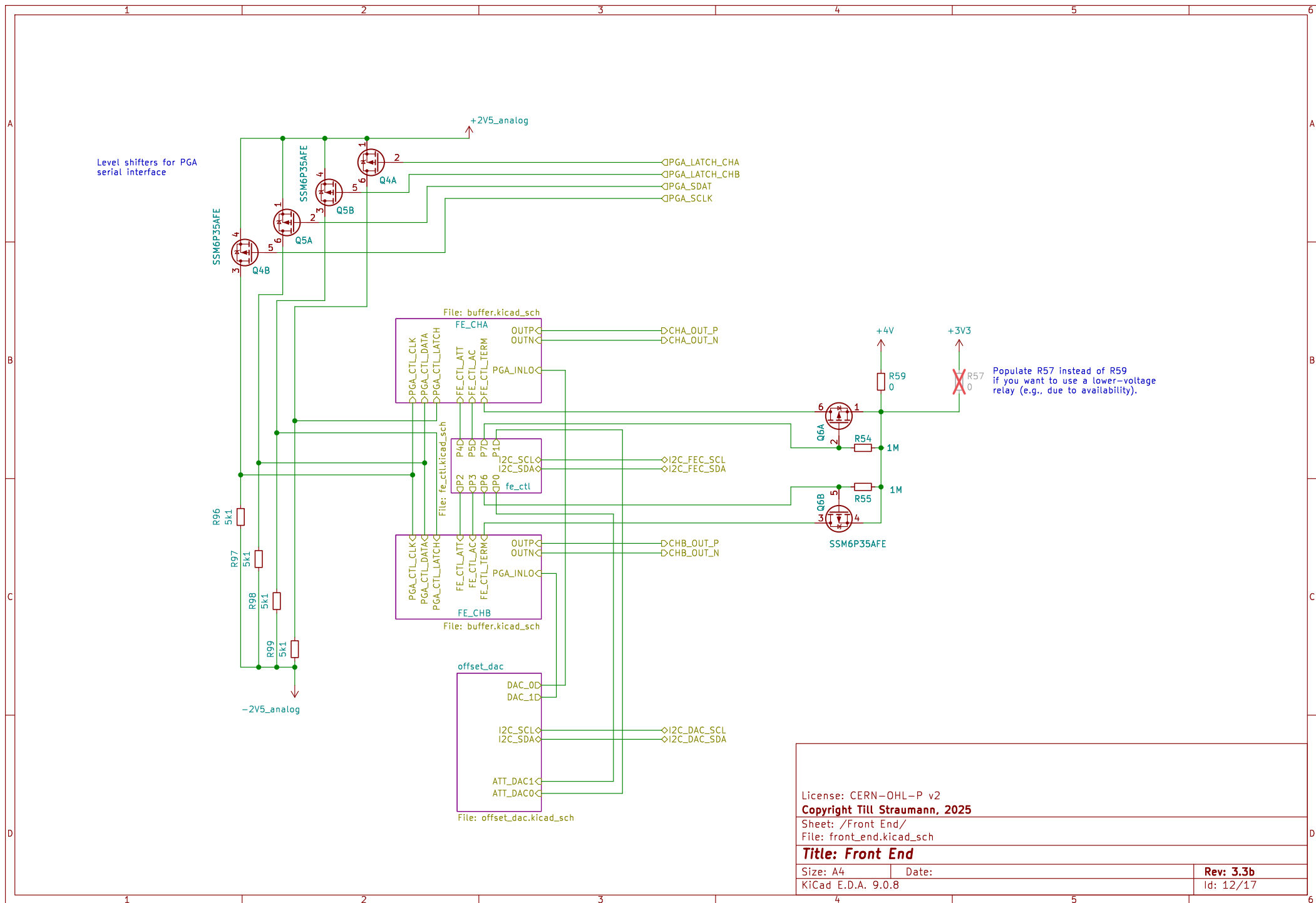
Title: USB Phy

Size: A4
 KiCad E.D.A. 9.0.8

Date:

Rev: 3.3b

Id: 11/17



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Sheet: /Front End/
 File: front_end.kicad_sch

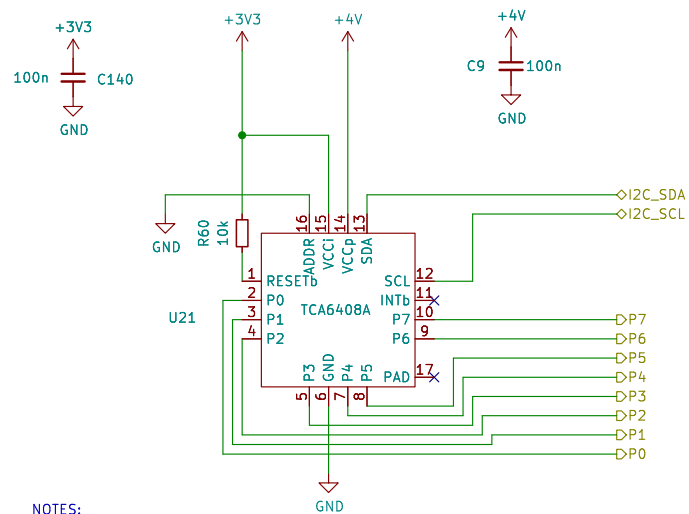
Title: Front End

Size: A4
 KiCad E.D.A. 9.0.8

Date:

Rev: 3.3b

Id: 12/17



NOTES:
 I2C Address 0x40
 ADDR: VCCp levels
 RESETb: VCCi levels
 PAD: seems unconnected

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Sheet: /Front End/fe_ctl/

File: fe_ctl.kicad_sch

Title: IO Extender / Level Shifter

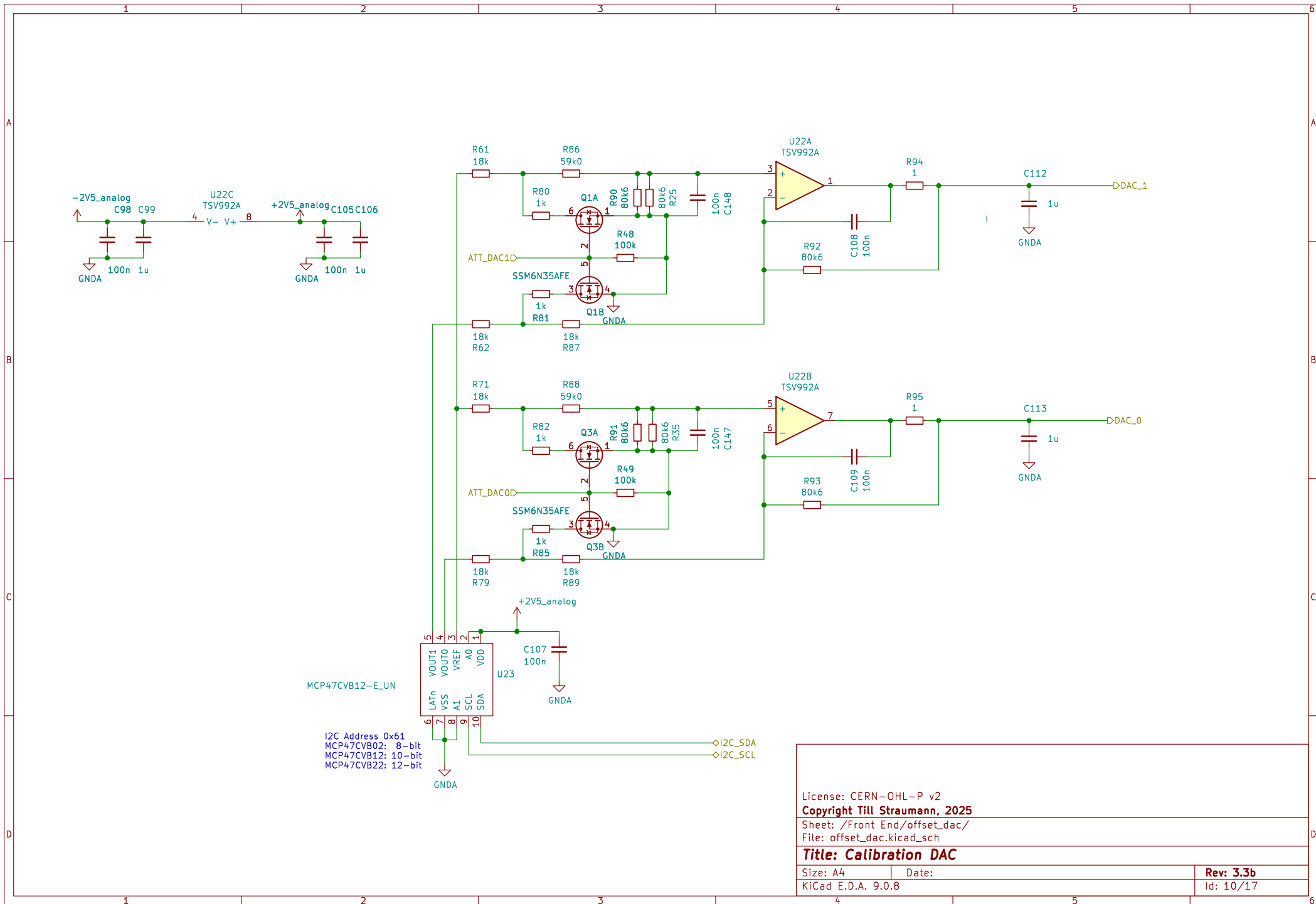
Size: A4

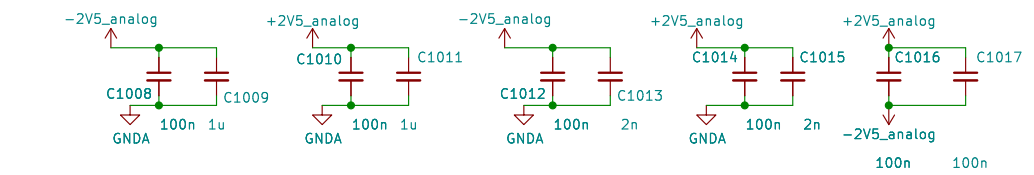
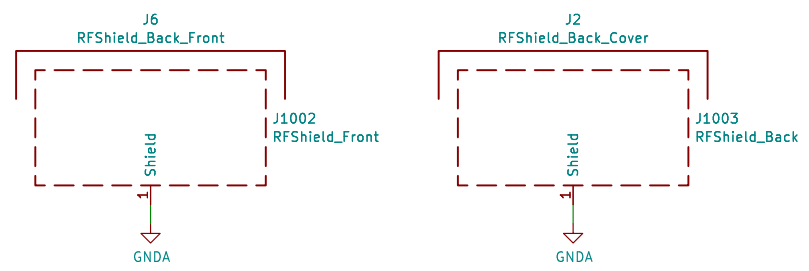
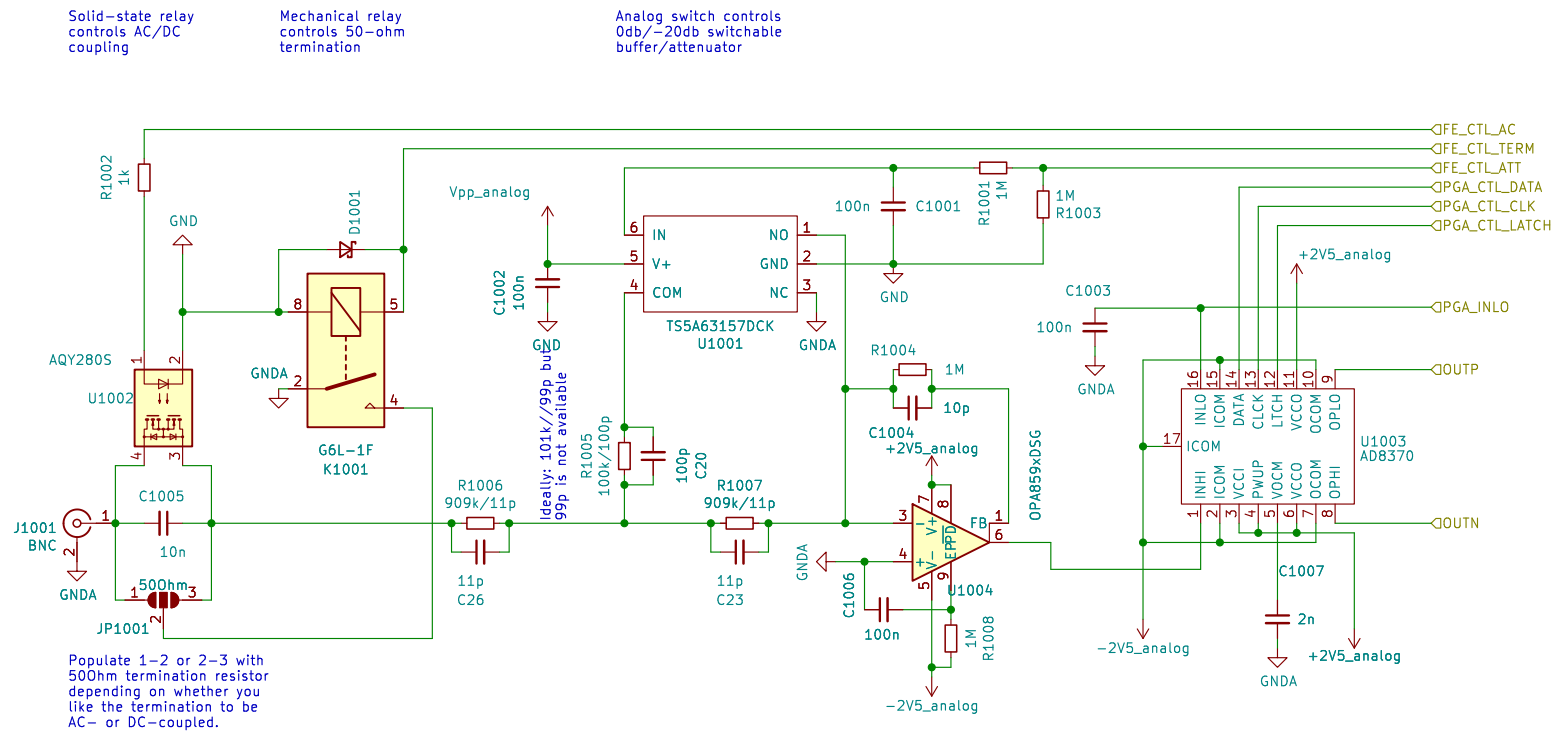
Date:

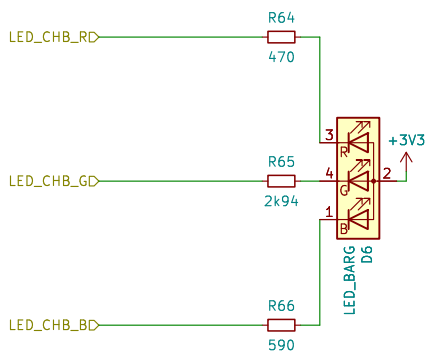
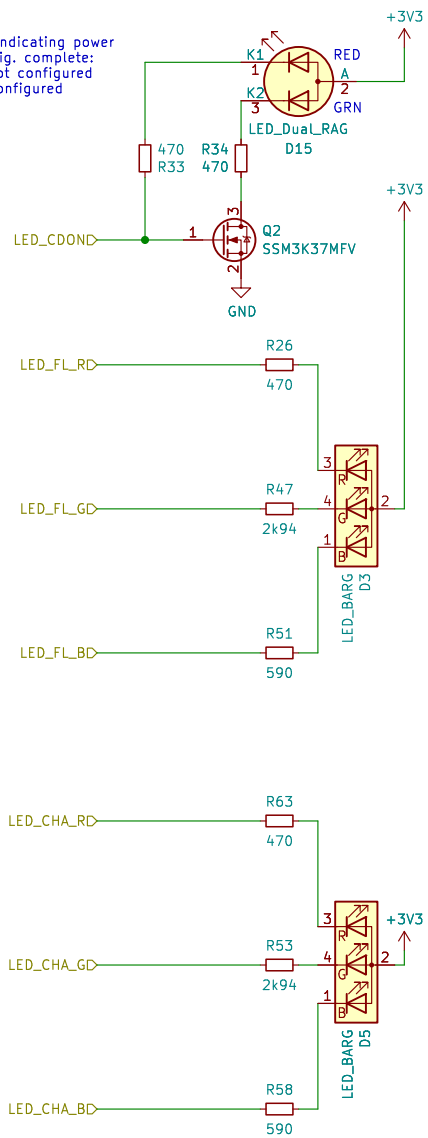
Rev: 3.3b

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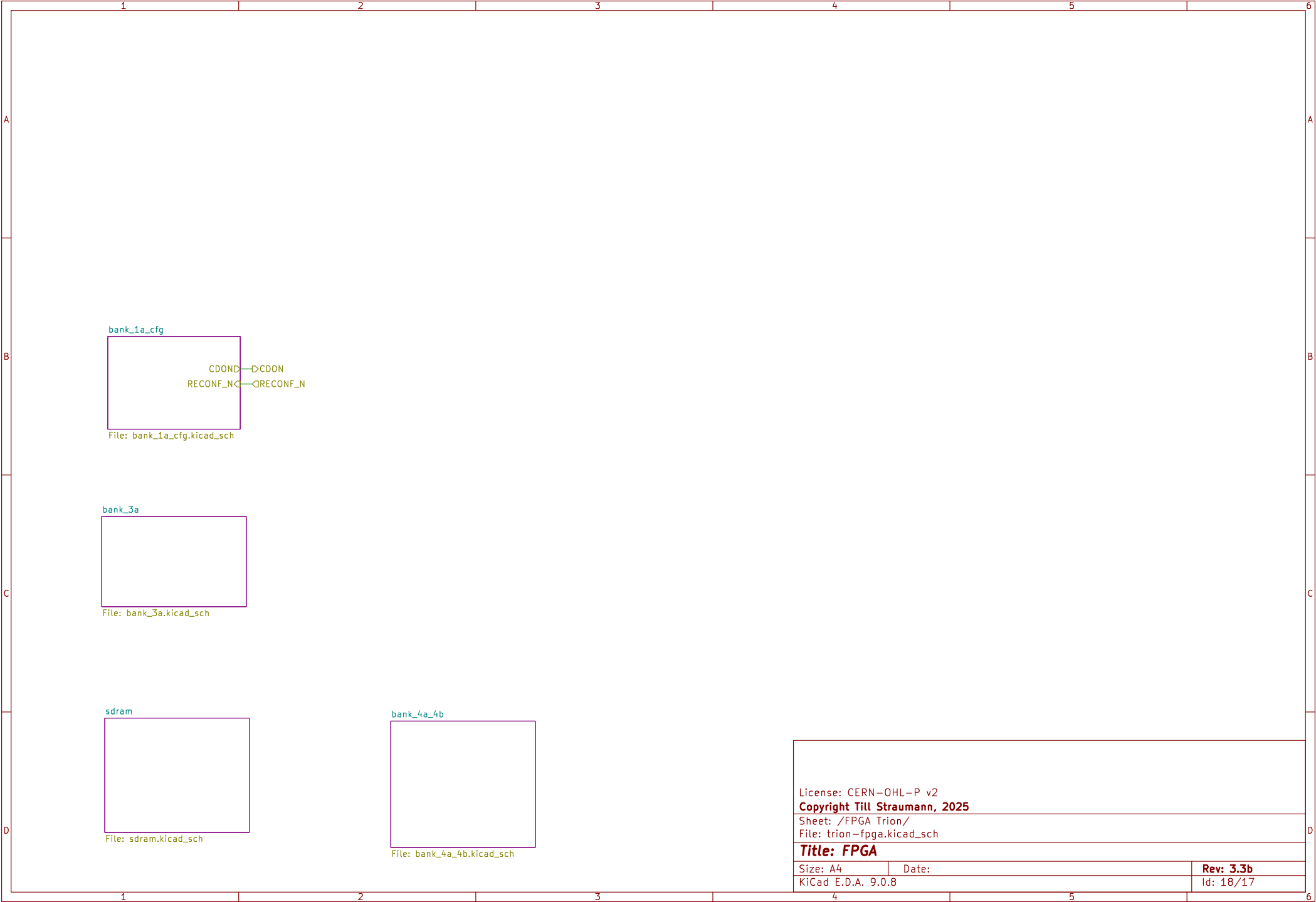
Id: 6/17







Id: 17/17

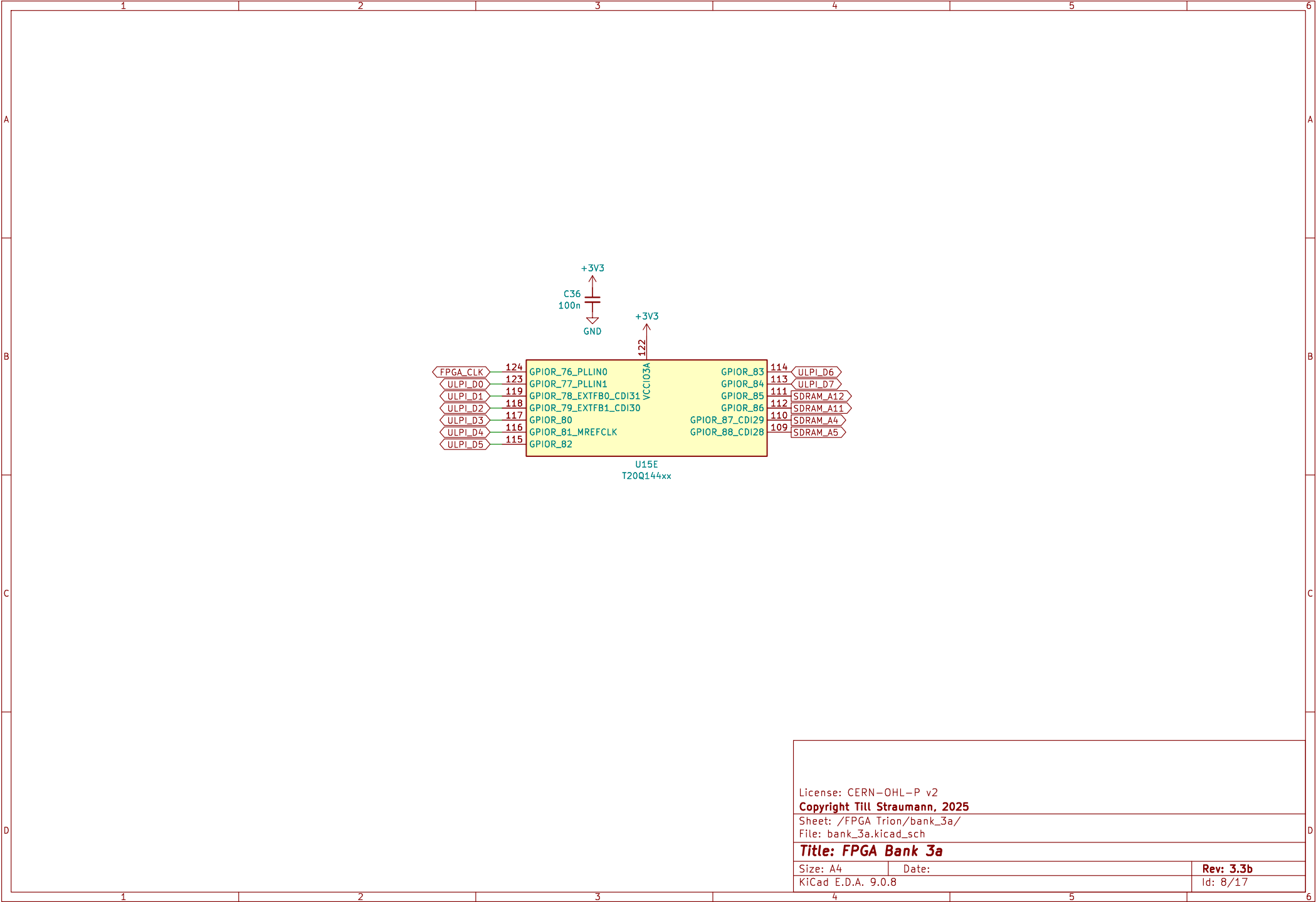


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Sheet: /FPGA Trion/
File: trion-fpga.kicad_sch

Title: FPGA

Size: A4	Date:	Rev: 3.3b
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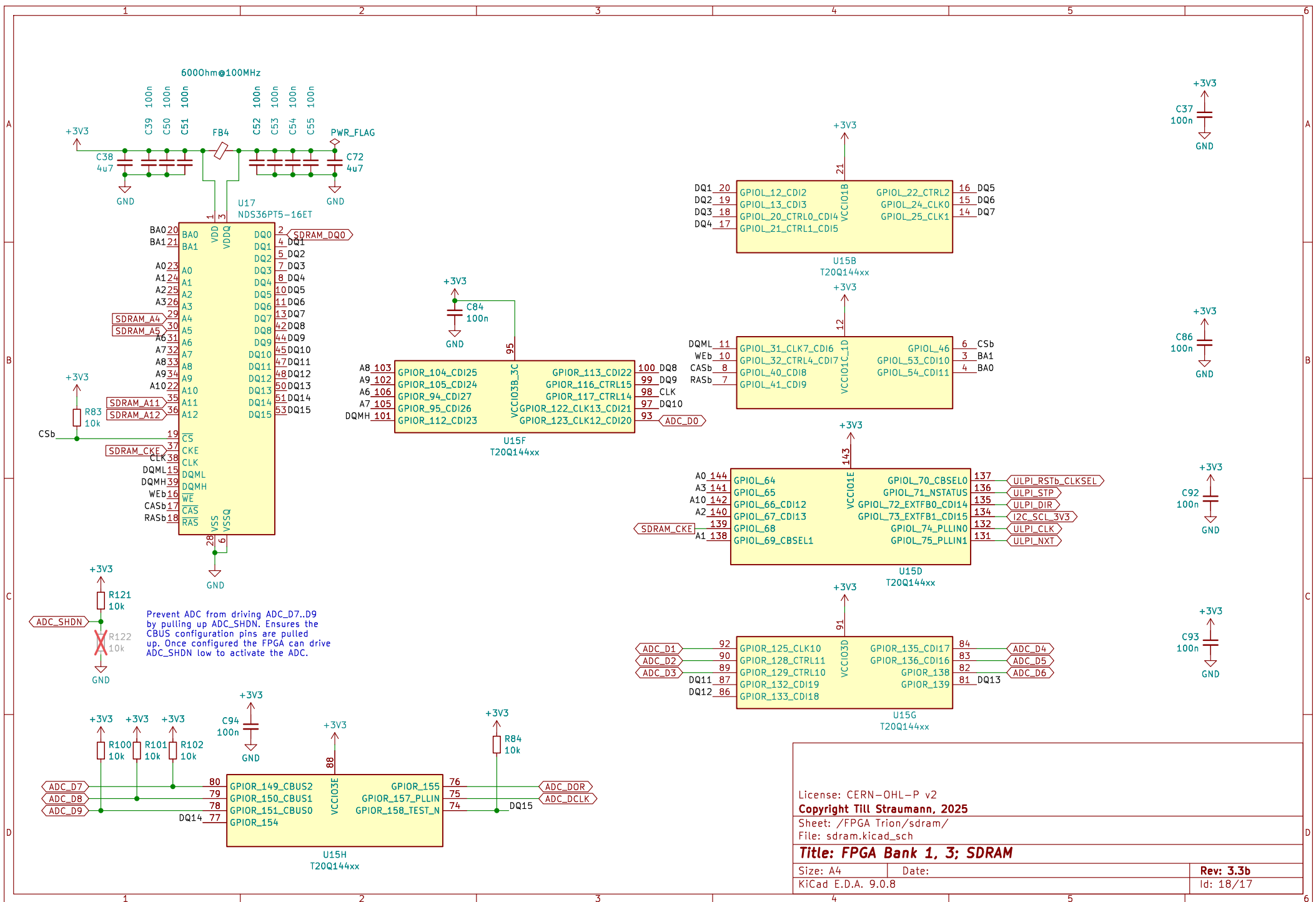


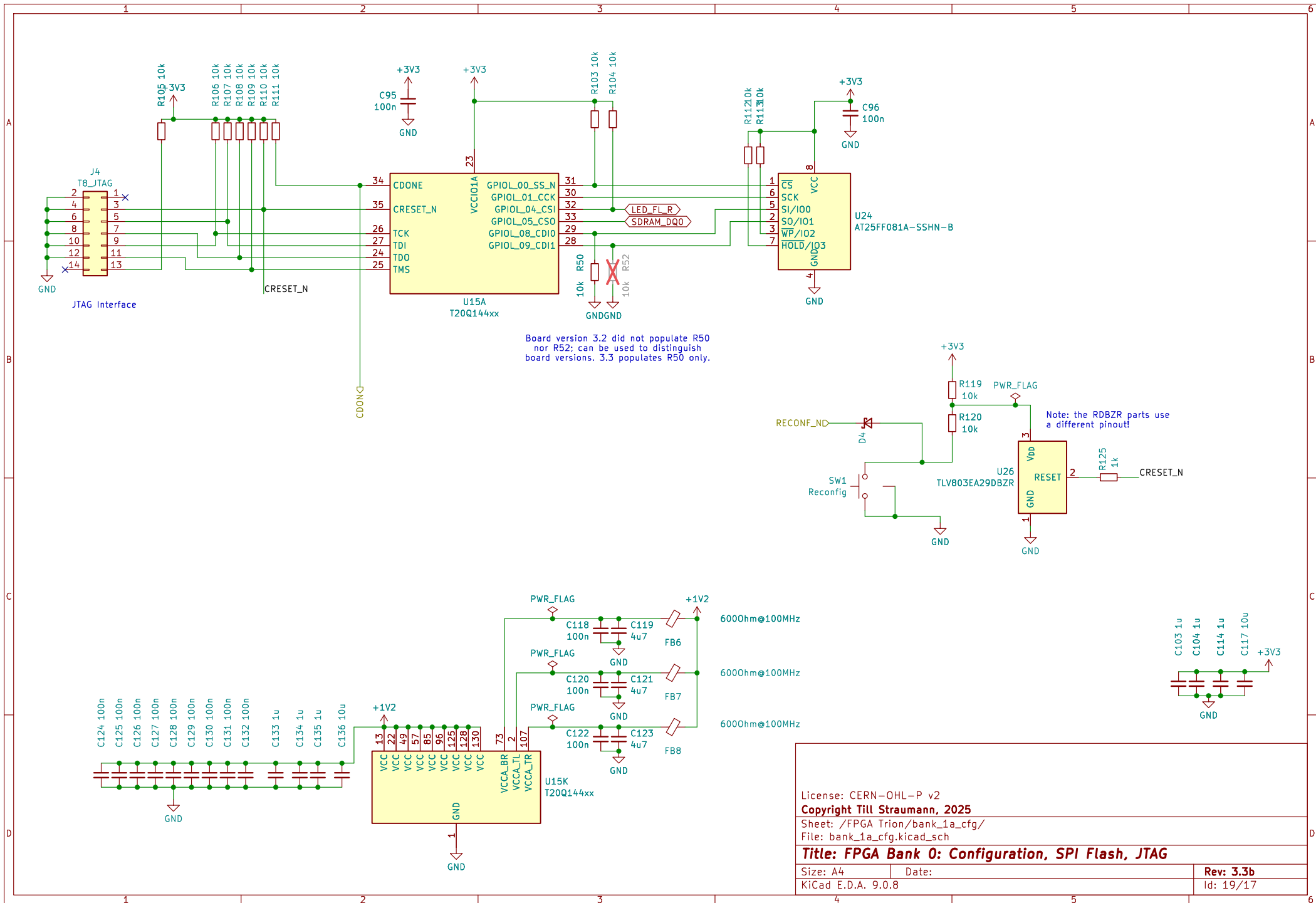
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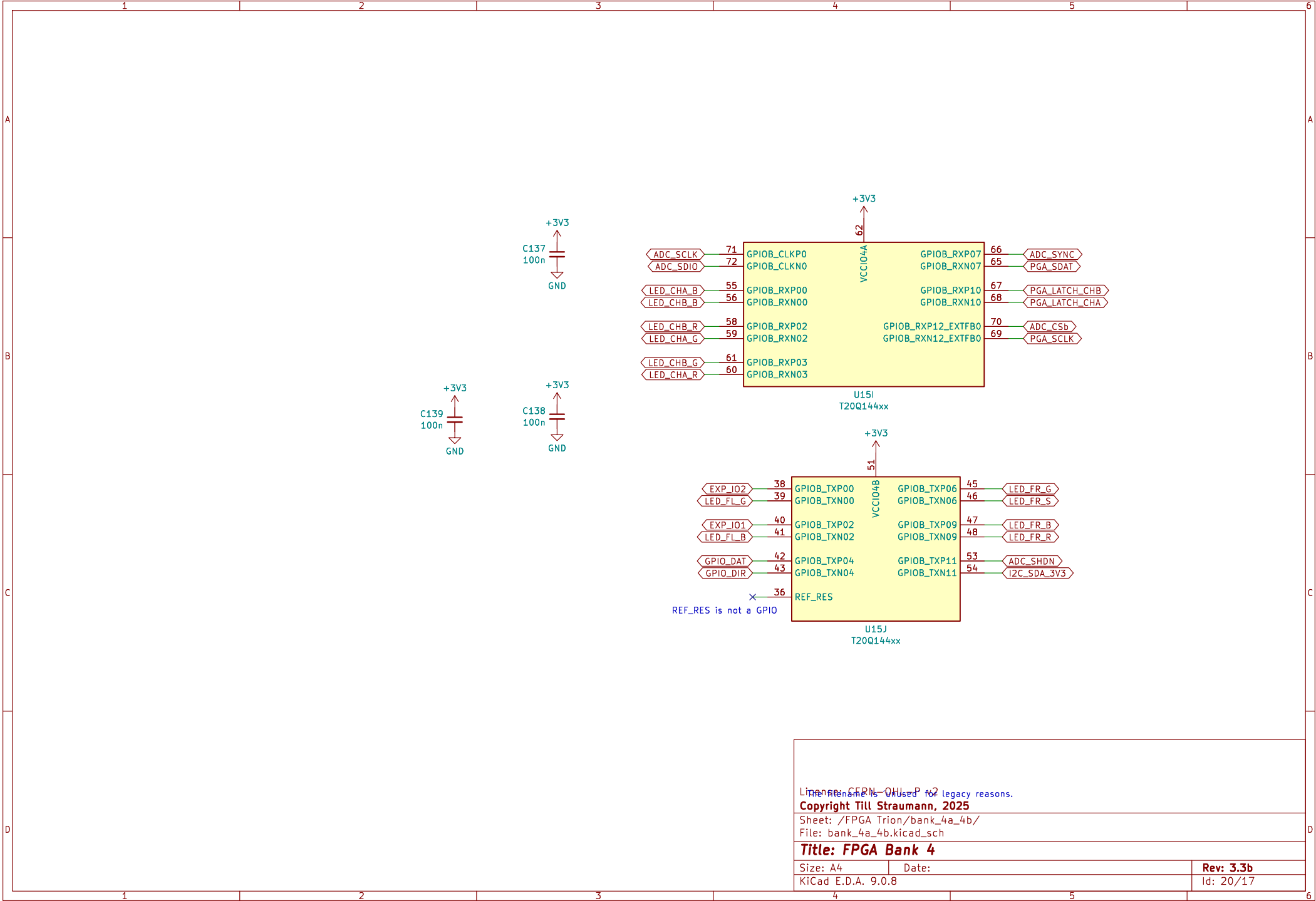
Sheet: /FPGA Trion/bank_3a/
File: bank_3a.kicad_sch

Title: FPGA Bank 3a

Size: A4	Date:	Rev: 3.3b
KiCad E.D.A. 9.0.8		Id: 8/17







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Sheet: /FPGA Trion/bank_4a_4b/

File: bank_4a_4b.kicad_sch

Title: FPGA Bank 4

Size: A4

Date:

KiCad E.D.A. 9.0.8

Rev: 3.3b

Id: 20/17